

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0236

Rec Harmony 345 kV
March 01, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0236. The Interconnection Customer proposes a 484.8 MW wind facility interconnecting at the Rec Harmony 345 kV substation in the West Virginia area. The request was received on May 16, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 152 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,398,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The North Adams / Rec Harmony 138kV is overloaded to 115.0% of its 276 MVA emergency rating for the loss of the Rec Harmony / Dillard-Orange 345kV Line. The project contributes 16.0% DFAX (77.3 MW) to this overload.

The Dillard-Orange / Rec Harmony 345kV is overloaded to 112.8% of its 1,309 MVA emergency rating for the loss of the Rec Harmony 345/138kV XFMR. The project contributes 69.8% DFAX (338.6 MW) to this overload.

Rec Harmony 345/138 kV Transformer Bank: 108.5% loading (771 MVA) following the outage of the Dillard-Orange / Rec Harmony 345kV Line; DFAX = 34.3%.

Burwell 500/345kV Auto: 101.3% loading (692 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 345-1 kV XFMR; DFAX = 8.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Adams / Rec Harmony 138kV	Rebuild line with higher-capacity conductor	\$3,000,000	24% (\$733,000)
Dillard-Orange / Rec Harmony 345kV	Rebuild line with higher-capacity conductor	\$65,534,000	24% (\$16,007,000)
Rec Harmony 345-138 kV XFMR	Replace transformer with higher-rated unit	\$15,947,000	24% (\$3,895,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Burwell 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$24,040,000	7% (\$1,763,000)
	Total allocated to project		\$22,398,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.